



MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of faculty: Sunil Kumar Sharma

Subject & Subject Code: AEM, 3CS1-01& 3AD1-01

Semester: III

Session: 2022-23(Odd Sem.)

Branch: CSE& AIDS

Batch: A, B & C

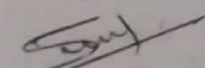
Unit No.1

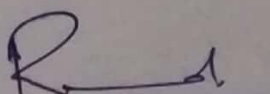
Date of Submission: 18-10-2022

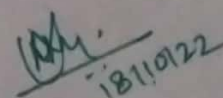
Lecture No.	Topics Covered	Page No.	Total Page
1	Discrete and Continuous Random Variables	1-8	8
2	Probability Distribution function	9-18	10
3	Mathematical Expectations	100-103	3
4	Moments	104-106	3
5	Moments Generating Function	107-109	3
6	Mean & Variance	110-112	3

References:

Chandrika Prasad, Advance Engineering Mathematics, Stanford Publishers and Distributors
Engineering Mathematics-T. Veerarajan, TMH


Faculty


HOD


18/10/22
DEAN ACDMICS

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

DISCRETE AND CONTINUOUS RANDOM VARIABLES

LECTURE NO: -1

Discrete and Continuous Random Variables:

What is a Random Variable?

A random variable is a variable that can take on many values. This is because there can be several outcomes of a random occurrence. Thus, a random variable should not be confused with an algebraic variable. An algebraic variable represents the value of an unknown quantity in an algebraic equation that can be calculated. On the other hand, a random variable can have a set of values that could be the resulting outcome of a random experiment.

Random Variable Definition

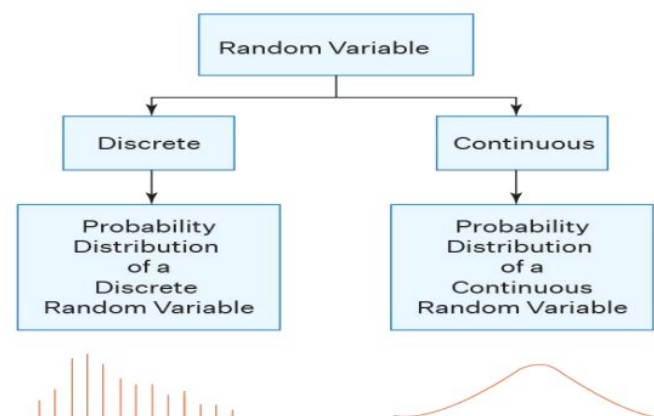
A random variable can be defined as a type of variable whose value depends upon the numerical outcomes of a certain random phenomenon. It is also known as a stochastic variable. Random variables are always real numbers as they are required to be measurable.

Random Variable Example

Suppose 2 dice are rolled and the random variable, X , is used to represent the sum of the numbers. Then, the smallest value of X will be equal to 2 ($1 + 1$), while the highest value would be 12 ($6 + 6$). Thus, X could take on any value between 2 to 12 (inclusive). Now if probabilities are attached to each outcome then the probability distribution of X can be determined.

Types of Random Variables

Types of a Random Variable



Random Variables can be divided into two broad categories depending upon the type of data available. These are given as follows:

- Discrete random variable
- Continuous random variable

A probability mass function is used to describe a discrete random variable and a probability density function describes a continuous random variable. The upcoming sections will cover these topics in detail.

Discrete Random Variable

A discrete random variable is a variable that can take on a finite number of distinct values. For example, the number of children in a family can be represented using a discrete random variable. A probability distribution is used to determine what values a random variable can take and how often does it take on these values. Some of the discrete random variables that are associated with certain special probability distributions will be detailed in the upcoming section.

Continuous Random Variable

A random variable that can take on an infinite number of possible values is known as a continuous random variable. Such a variable is defined over an interval of values rather than a specific value. An example of a continuous random variable is the weight of a person. The probability that a continuous random variable takes on an exact value is 0 thus, a probability density function is used to describe such a variable. Some commonly used continuous random variables are given below.

Mean of a Random Variable

The average value of a random variable is called the mean of a random variable. The mean is also known as the expected value. It is generally denoted by $E[X]$, where X is the random variable. The mean or expected value of a random variable can also be defined as the weighted average of all the values of the variable. The formulas for the mean of a random variable are given below:

- Mean of a Discrete Random Variable: $E[X] = \sum xP(X=x)$. Here $P(X = x)$ is the probability mass function.
- Mean of a Continuous Random Variable: $E[X] = \int xf(x)dx$. $f(x)$ is the probability density function

Variance of a Random Variable

The variance of a random variable can be defined as the expected value of the square of the difference of the random variable from the mean. The variance of a random variable is given by $\text{Var}[X]$ or σ^2 . If μ is the mean then the formula for the variance is given as follows:

- Variance of a Discrete Random Variable: $\text{Var}[X] = \sum (x-\mu)^2 P(X=x)$
- Variance of a Continuous Random Variable: $\text{Var}[X] = \int (x-\mu)^2 f(x)dx$

A **discrete variable** is a variable whose value is obtained by counting.

Examples: number of students present
 number of red marbles in a jar
 number of heads when flipping three coins
 students' grade level

A **discrete random variable** X has a countable number of possible values.

Example: Let X represent the sum of two dice.

Then the probability distribution of X is as follows:

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

To graph the probability distribution of a discrete random variable, construct a **probability histogram**.

A **continuous variable** is a variable whose value is obtained by measuring.

Examples: height of students in class
 weight of students in class
 time it takes to get to school
 distance traveled between classes

A **continuous random variable** X takes all values in a given interval of numbers.

- The probability distribution of a continuous random variable is shown by a **density curve**.
- The probability that X is between an interval of numbers is the area under the density curve between the interval endpoints
- The probability that a **continuous random variable** X is exactly equal to a number is zero

Continuous Random Variable vs Discrete Random Variable

Both discrete and continuous random variables are used to model a random phenomenon. The differences between a continuous random variable and **discrete random variable** are given in the table below:

Continuous Random Variable	Discrete Random Variable
The value of a continuous random variable falls between a range of values.	The value of a discrete random variable is an exact value.
The probability density function is associated with a continuous random variable.	The probability mass function is used to describe a discrete random variable
A continuous random variable can take on an infinite number of values.	Such a variable can take on a finite number of distinct values.
Mean of a continuous random variable is $E[X] = \int_{-\infty}^{\infty} xf(x)dx$	The mean of a discrete random variable is $E[X] = \sum x P(X = x)$, where $P(X = x)$ is the probability mass function.
The variance of a continuous random variable is $\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$	The variance of a discrete random variable is $\text{Var}[X] = \sum (x - \mu)^2 P(X = x)$
The examples of a continuous random variable are uniform random variable, exponential random variable, normal random variable, and standard normal random variable.	The examples of a discrete random variable are binomial random variable, geometric random variable, Bernoulli random variable, and Poisson random variable.

Examples on Random Variable

Suppose that to each point of a sample space we assign a number. We then have a *function* defined on the sample space. This function is called a *random variable* (or *stochastic variable*) or more precisely a *random function* (*stochastic function*). It is usually denoted by a capital letter such as X or Y . In general, a random variable has some specified physical, geometrical, or other significance.

EXAMPLE Suppose that a coin is tossed twice so that the sample space is $S = \{HH, HT, TH, TT\}$. Let X represent the number of heads that can come up. With each sample point we can associate a number for X as shown in Table 2-1. Thus, for example, in the case of HH (i.e., 2 heads), $X = 2$ while for TH (1 head), $X = 1$. It follows that X is a random variable.

Sample Point	$\begin{matrix} H \\ H \end{matrix}$	HT	$\begin{matrix} T \\ H \end{matrix}$	TT
X	2	1	1	0

It should be noted that many other random variables could also be defined on this sample space, for example, the square of the number of heads or the number of heads minus the number of tails.

A random variable that takes on a finite or countably infinite number of values is called a *discrete random variable* while one which takes on a non-countably infinite number of values is called a *non-discrete random variable*.

Discrete Distribution Function

Let X be a discrete random variable, and suppose that the possible values that it can assume are given by $x_1, x_2, x_3, \dots$, arranged in some order. Suppose also that these values are assumed with probabilities given by

$$P(X = x_k) = f(x_k) \quad k = 1, 2, \dots \quad (1)$$

It is convenient to introduce the *probability function*, also referred to as *probability distribution*, given by

$$P(X = x) = f(x) \quad (2)$$

For $x = x_k$, this reduces to (1) while for other values of x , $f(x) = 0$.

In general, $f(x)$ is a probability function if

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$

where the sum in 2 is taken over all possible values of x .

the probability function is thus given by Table

x	0	1	2
$f(x)$	$1/4$	$1/2$	$1/4$

Distribution Functions for Random Variables

The cumulative distribution function, or briefly the distribution function, for a random variable X is defined by

$$F(x) = P(X \leq x) \quad (3)$$

where x is any real number, i.e., $-\infty < x < \infty$.

The distribution function $F(x)$ has the following properties:

1. $F(x)$ is nondecreasing [i.e., $F(x) < F(y)$ if $x < y$].
2. $\lim_{x \rightarrow -\infty} F(x) = 0$; $\lim_{x \rightarrow \infty} F(x) = 1$.
3. $F(x)$ is continuous from the right [i.e., $\lim_{h \rightarrow 0^+} F(x+h) = F(x)$ for all x].

Distribution Functions for Discrete Random Variable

The distribution function for a discrete random variable X can be obtained from its probability function by noting that, for all x in $(-\infty, \infty)$,

$$F(x) = P(X \leq x) = \sum_{u \leq x} f(u)$$

where the sum is taken over all values u taken on by X for which $u \leq x$.

If X takes on only a finite number of values $x_1, x_2, \dots, x_n$, then the distribution function is given by

$$F(x) = \begin{cases} 0 & -\infty < x < x_1 \\ f(x_1) & x_1 \leq x < x_2 \\ f(x_1) + f(x_2) & x_2 \leq x < x_3 \\ \vdots & \vdots \\ f(x_1) + \dots + f(x_n) & x_n \leq x < \infty \end{cases}$$

EXAMPLE 2.3 Suppose that a coin is tossed twice so that the sample space is $S = \{HH, HT, TH, TT\}$. Let X represent the number of heads that can come up. With each sample point we can associate a number for X as shown in Table. Thus, for example, in the case of HH (i.e., 2 heads), $X = 2$ while for TH (1 head), $X = 1$. It follows that X is a random variable.

Sample	H	HT	T	TT
--------	-----	------	-----	------

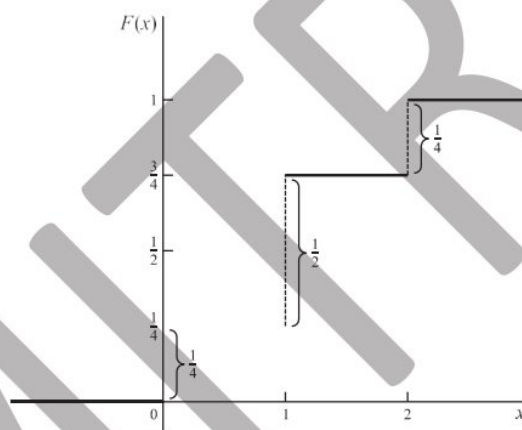
Point	H		H	
X	2	1	1	0

(a) Find the distribution function for the random variable X of (b) Obtain its graph.

(a) The distribution function is

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & 2 \leq x < \infty \end{cases}$$

(b) The graph of $F(x)$ is shown in Fig.



The following things about the above distribution function, which are true in general, should be noted.

1. The magnitudes of the jumps at 0, 1, 2 are $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ which are precisely the probabilities in Table 2-2. This fact

$$\frac{1}{4} \quad \frac{1}{2} \quad \frac{1}{4}$$

enables one to obtain the probability function from the distribution function.

2. Because of the appearance of the graph of Fig. 2-1, it is often called a *staircase function* or *step function*. The value of the function at an integer is obtained from the higher step; thus the value at 1 is $\frac{3}{4}$ and not $\frac{1}{4}$. This

$$\frac{3}{4} \quad \frac{1}{4}$$

is expressed mathematically by stating that the distribution function is *continuous from the right* at 0, 1, 2.

3. As we proceed from left to right (i.e. going *upstairs*), the distribution function either remains the same or increases, taking on values from 0 to 1. Because of this, it is said to

be a *monotonically increasing function*.

It is clear from the above remarks and the properties of distribution functions that the probability function of a discrete random variable can be obtained from the distribution function by noting that

$$f(x) = F(x) - \lim_{u \rightarrow x^-} F(u)$$

Continuous Random Variables

A non-discrete random variable X is said to be *absolutely continuous*, or simply *continuous*, if its distribution function may be represented as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

$$-\infty < x < \infty$$

where the function $f(x)$ has the properties

$$1. f(x) \geq 0$$

$$2. \int_{-\infty}^{\infty} f(x) dx = 1$$

It follows from the above that if X is a continuous random variable, then the probability that X takes on any one particular value is zero, whereas the interval probability that X lies between two different values, say, a and b , is given by

$$P(a < x < b) = \int_a^b f(x) dx$$

References:

Engineering Mathematics, T Veerarajan, TMH

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

PROBABILITY DISTRIBUTION FUNCTION

LECTURE NO: -2

Probability Distributions Function

A random variable can take multiple values. One very important thing is to know if some values will be more often encountered than others. The description of the probability of each possible value that a random variable can take is called its **probability distribution**.

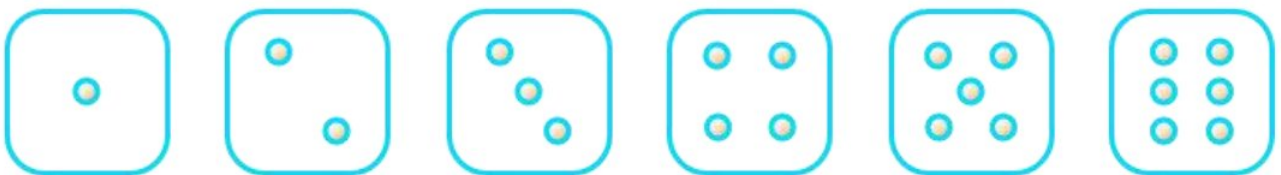
This idea of discrete versus continuous variable is important and we will study the concept of probability distribution in both cases. Even if they are related, some differences exist.

Anyway, the probability distribution of a random variable x describes the probability of each outcome (a probability of 1 meaning that the variable will always take this value and a probability of 0 that it will never be encountered). This function is called probability distribution. More specifically, it is called the **probability mass function** for a discrete variable and **probability density function** for a continuous variable.

Discrete Variable and Probability Mass Function

The **probability mass function** is the function which describes the probability associated with the random variable x . This function is named $P(x)$ or $P(x=x)$ to avoid confusion. $P(x=x)$ corresponds to the probability that the random variable x take the value x (note the different typefaces).

Example



Dice experiment illustrating discrete random variable and probability mass function

Let's roll a die an infinite number of times and look at the proportion of 1, the proportion of 2 and so on. We call x the random variable that corresponds to the outcome of the dice roll. Thus the random variable x can only take the following discrete values: 1, 2, 3, 4, 5 or 6. It is thus a **discrete random variable**.

The aim of the probability mass function is to describe the probability of each possible value. In our example, it describes the probability to get a 1, the probability to get a 2 and

so on. In the case of a dice rolling experiment, we have the same probability to get each value (if we assume that the die is perfect).

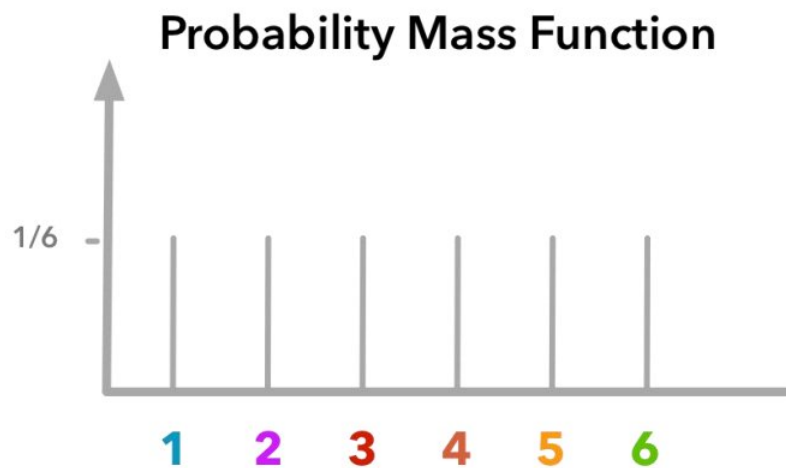
Now, how can we calculate the probabilities $P(x=1)$, $P(x=2)$ etc.? Since we have 6 possible outcomes and that they are equiprobable we have:

So a random variable can take multiple values. One very important thing is to know if some values will be more often encountered than others. The description of the probability of each possible value that a random variable can take is called its **probability distribution**.

This idea of discrete versus continuous variable is important and we will study the concept of probability distribution in both cases. Even if they are related, some differences exist.

Anyway, the probability distribution of a random variable x describes the probability of each outcome (a probability of 1 meaning that the variable will always take this value and a probability of 0 that it will never be encountered). This function is called probability distribution. More specifically, it is called the **probability mass function** for a discrete variable and **probability density function** for a continuous variable.

The probability mass function could look something like:



Probability mass function of the dice experiment

The y-axis gives the probability and x-axis the outcome.

The **Probability Mass Function (PMF)** is also called a **probability function** or **frequency function** which characterizes the distribution of a discrete random variable. Let X be a discrete random variable of a function, then the probability mass function of a **random variable** X is given by

$P_X(x) = P(X=x)$, For all x belongs to the range of X

It is noted that the probability function should fall on the condition :

- $P_x(x) \geq 0$ and
- $\sum_{x \in \text{Range}(x)} P_x(x) = 1$

Here the $\text{Range}(X)$ is a countable set and it can be written as $\{x_1, x_2, x_3, \dots\}$. This means that the random variable X takes the value $x_1, x_2, x_3, \dots$

These can also be stated as explained below.

The probability mass function $P(X = x) = f(x)$ of a discrete random variable is a function that satisfies the following properties:

- $P(X = x) = f(x) > 0$; if $x \in \text{Range of } x \text{ that supports}$
- $\sum_{x \in \text{Range}(x)} P_x(x) = 1$



Definition

The Probability Mass function is defined on all the values of R , where it takes all the arguments of any **real number**. It doesn't belong to the value of X when the argument value equals to zero and when the argument belongs to x , the value of PMF should be positive.

The probability mass function is usually the primary component of defining a discrete probability distribution, but it differs from the **probability density function (PDF)** where it produces distinct outcomes. This is the reason why probability mass function is used in computer programming and statistical modelling. In other words, probability mass function is a function that relates discrete events to the probabilities associated with those events occurring. The word "mass" indicates the probabilities that are concentrated on discrete events.

Continuous Variable and Probability Density Function

Some variables are not discrete. They can take an infinite number of values in a certain range. But we still need to describe the probability associated with outcomes. The equivalent of the probability mass function for a continuous variable is called the **probability density function**.

In the case of the probability mass function, we saw that the y-axis gives a probability. For instance, in the plot we created with Python, the probability to get a 1 was equal to $1/6 \approx 0.16$ (check on the plot above). It is $1/6$ because it is one possibility over 6 total possibilities.

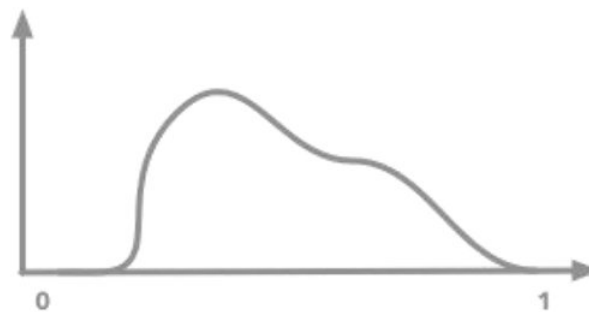
However, we can't do this for continuous variables because the total number of possibilities is infinite. For instance, if we draw a number between 0 and 1, we have an infinite number of possible outcomes (for instance 0.320502304...). In the example above,

we had 6 possible outcomes, leading to probabilities around $1/6$. Now, we have each probability equal to $1/+\infty \approx 0$. Such a function would not be very useful.

For that reason, the y-axis of the probability density function doesn't represent probability values. To get the probability, we need to calculate the **area under the curve** (we will see below some details about the area under the curve). The advantage is that it leads to the probabilities according to a certain range (on the x-axis): the area under the curve increases if the range increases. Let's see some examples to clarify all of this.

Example 4.

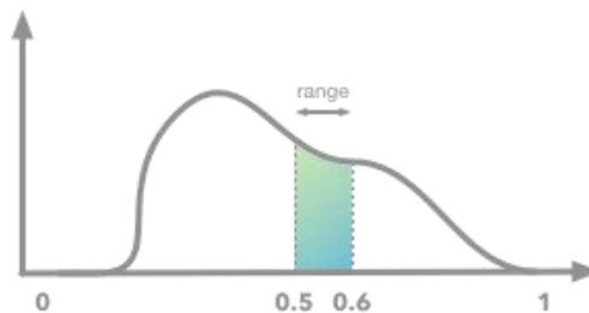
Let's say that we have a random variable x that can take values between 0 and 1. Here is its probability density function:



Probability density function

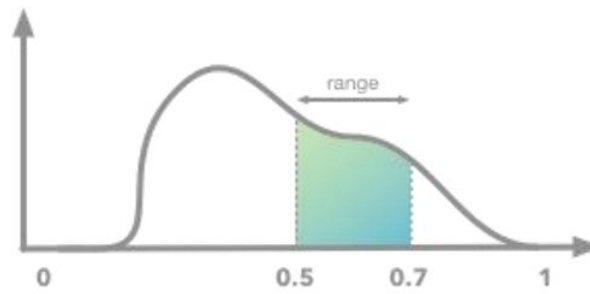
We can see that 0 seems to be not possible (probability around 0) and neither 1. The pic around 0.3 means that will get a lot of outcomes around this value.

Finding probabilities from probability density function between a certain range of values can be done by calculating the area under the curve for this range. For example, the probability of drawing a value between 0.5 and 0.6 corresponds to the following area:



Probability density function and area under the curve between 0.5 and 0.6.

We can easily see that if we increase the range, the probability (the area under the curve) will increase as well. For instance, for the range of 0.5-0.7:



Probability density function and area under the curve between 0.5 and 0.7.

We will see in a moment how to calculate the area under the curve and get the probability associated with a specific range.

By the way, this distribution shows the same probability for each value: it is called the **uniform distribution**.

Joint probability distribution

Now let's see what happens if we roll two dice. For each die, the outcomes are associated with a certain probability. We need two random variables to describe the game, let's say that x corresponds to the first die and y to the second one. We also have two probability mass functions associated with the random variables: $P(x)$ and $P(y)$. Here the possible values of the random variables (1, 2, 3, 4, 5 or 6) and the probability mass functions are actually the same for both dice, but it doesn't need to be the case.

The **joint probability distribution** is useful in the cases where we are interested in the probability that x takes a specific value while y takes another specific value. For instance, what would be the probability to get a 1 with the first dice and 2 with the second dice? The probabilities corresponding to every pair of values are written $P(x=x, y=y)$ or $P(x, y)$. This is what we call the **joint probability**.

Continuous Variable and Probability Density Function

Some variables are not discrete. They can take an infinite number of values in a certain range. But we still need to describe the probability associated with outcomes. The equivalent of the probability mass function for a continuous variable is called the **probability density function**.

In the case of the probability mass function, we saw that the y-axis gives a probability. For instance, in the plot we created with Python, the probability to get a 1 was equal to $1/6 \approx 0.16$ (check on the plot above). It is $1/6$ because it is one possibility over 6 total possibilities.

However, we can't do this for continuous variables because the total number of possibilities is infinite. For instance, if we draw a number between 0 and 1, we have an infinite number of possible outcomes (for instance 0.320502304...). In the example above, we had 6 possible outcomes, leading to probabilities around $1/6$. Now, we have each probability equal to $1/\infty \approx 0$. Such a function would not be very useful.

For that reason, the y-axis of the probability density function doesn't represent probability values. To get the probability, we need to calculate the **area under the curve** (we will see below some details about the area under the curve). The advantage is that it leads to the probabilities according to a certain range (on the x-axis): the area under the curve increases if the range increases. Let's see some examples to clarify all of this.

What is the Probability Density Function?

The Probability Density Function(PDF) defines the probability function representing the density of a continuous random variable lying between a specific range of values. In other words, the probability density function produces the likelihood of values of the continuous random variable. Sometimes it is also called a **probability distribution** function or just a probability function. However, this function is stated in many other sources as the function over a broad set of values. Often it is referred to as cumulative distribution function or sometimes as **probability mass function**(PMF). However, the actual truth is PDF (probability density function) is defined for continuous random variables, whereas PMF (probability mass function) is defined for discrete random variables.

Probability Density Function Formula

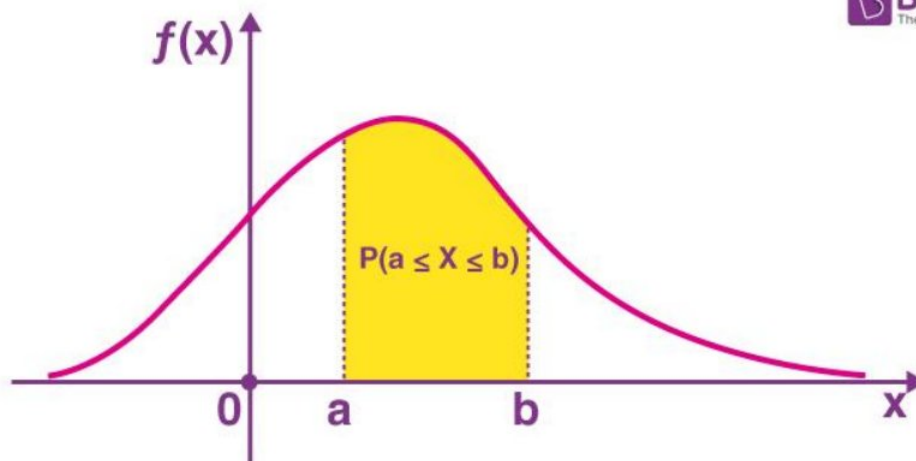
In the case of a continuous random variable, the probability taken by X on some given value x is always 0. In this case, if we find $P(X = x)$, it does not work. Instead of this, we must calculate the probability of X lying in an interval (a, b). Now, we have to figure it for $P(a < X < b)$, and we can calculate this using the formula of PDF. The Probability density function formula is given as,

Or

This is because, when X is continuous, we can ignore the endpoints of intervals while finding probabilities of continuous random variables. That means, for any constants a and b, $P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$.

Probability Density Function Graph

The probability density function is defined as an integral of the density of the variable density over a given range. It is denoted by $f(x)$. This function is positive or non-negative at any point of the graph, and the integral, more specifically the **definite integral** of PDF over the entire space is always equal to one. The graph of PDFs typically resembles a bell curve, with the probability of the outcomes below the curve. The below figure depicts the graph of a probability density function for a continuous random variable x with function $f(x)$.



Probability Density Function Properties

Let x be the continuous random variable with density function $f(x)$, and the probability density function should satisfy the following conditions:

- For a continuous random variable that takes some value between certain limits, say a and b , the PDF is calculated by finding the area under its curve and the X -axis within the lower limit (a) and upper limit (b). Thus, the PDF is given by
- The probability density function is non-negative for all the possible values, i.e. $f(x) \geq 0$, for all x .
- The area between the density curve and horizontal X -axis is equal to 1, i.e.
- Due to the property of continuous random variables, the density function curve is **continued** for all over the given range. Also, this defines itself over a range of continuous values or the domain of the variable.

Probability Density Function Example

Question:

Let X be a continuous random variable with the PDF given by:

Find $P(0.5 < x < 1.5)$.

Solution:

Given PDF is:

Let us split the integral by taking the intervals as given below:

Substituting the corresponding values of $f(x)$ based on the intervals, we get;

Integrating the functions, we get;

$$= [(1)2/2 - (0.5)2/2] + \{[2(1.5) - (1.5)2/2] - [2(1) - (1)2/2]\}$$

$$= [(\frac{1}{2}) - (\frac{1}{8})] + \{[3 - (\frac{9}{8})] - [2 - (\frac{1}{2})]\}$$

$$= (\frac{3}{8}) + [(15/8) - (3/2)]$$

$$= (3 + 15 - 12)/8$$

$$= 6/8$$

$$= 3/4$$

Applications of Probability Density Function

PDF, i.e. the probability density function has many applications in different fields of study such as statistics, science and engineering. Some of the important applications of the probability density function are listed below:

- In Statistics, it is used to calculate the probabilities associated with the random variables.
- The probability density function is used in modelling the annual data of atmospheric NOx temporal concentration
- It is used to model the diesel engine combustion.

Joint Distribution

The above ideas are easily generalized to two or more random variables. We consider the typical case of two random variables that are either both discrete or both continuous. In cases where one variable is discrete and the other continuous, appropriate modifications are easily made. Generalizations to more than two variables can also be made.

1. DISCRETE CASE. If X and Y are two discrete random variables, we define the joint probability function of X and Y by

$$P(X = x, Y = y) = f(x, y)$$

where

1. $f(x, y) \geq 0$

2. $\sum_X \sum_Y f(X, Y) = 1$

i.e., the sum over all values of x and y is 1.

Suppose that X can assume any one of m values $x_1, x_2, \dots, x_m$ and Y can assume any one of n values $y_1, y_2, \dots, y_n$. Then the probability of the event that $X = x_j$ and $Y = y_k$ is given by $P(X = x_j, Y = y_k) = f(x_j, y_k)$

A joint probability function for X and Y can be represented by a joint probability table. The probability that $X = x_j$ is obtained by adding all entries in the row corresponding to x_j and is given by

$$P(X = x_j) = f_1(x_j) = \sum_{k=1}^n f(x_j, y_k)$$

X	Y	Y_1	Y_2		Y_n	Total
X_1		$f(x_1, y_1)$	$f(x_1, y_2)$		$f(x_1, y_n)$	$F_1(x_1)$
X		$f(x_2, y_1)$	$f(x_2, y_2)$		$f(x_2, y_n)$	$F(x_2)$
.....						
X_m		$f(x_m, y_1)$	$f(x_m, y_2)$		$f(x_m, y_n)$	$F(x_m)$
Total		$F_2(y_1)$	$F_2(y_2)$		$F_2(y_n)$	1

For $j = 1, 2, \dots, m$, these are indicated by the entry totals in the extreme right-hand

column or margin of Table 2-3. Similarly the probability that $Y = y_k$ is obtained by adding all entries in the column corresponding to y_k and is given by

$$P(Y = y_k) = f_2(y_k) = \sum_{j=1}^m f(x_j, y_k)$$

For $k = 1, 2, \dots, n$, these are indicated by the entry totals in the bottom row or margin of Table 2-3. Because the probabilities (15) and (16) are obtained from the margins of the table, we often refer to $f_1(x_j)$ and $f_2(y_k)$ [or simply $f_1(x)$ and $f_2(y)$] as the marginal probability functions of X and Y , respectively

It should also be noted that

$$\sum_{j=1}^m f_1(x_j) = \sum_{k=1}^n f_2(y_k) = 1$$

which can be written

$$\sum_{j=1}^m \sum_{k=1}^n f(x_j, y_k) = 1$$

This is simply the statement that the total probability of all entries is 1. The grand total of 1 is indicated in the lower right-hand corner of the table. The joint distribution function of X and Y is defined by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{u \leq x} \sum_{v \leq y} f(u, v)$$

In Table, $F(x, y)$ is the sum of all entries for which $x_j \leq x$ and $y_k \leq y$.

2. CONTINUOUS CASE. The case where both variables are continuous is obtained easily by analogy with the discrete case on replacing sums by integrals. Thus the joint probability function for the random variables X and Y (or, as it is more commonly called, the joint density function of X and Y) is defined by

$$1. f(x, y) \geq 0$$

$$2. \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Graphically $z = f(x, y)$ represents a surface, called the probability surface, as indicated in Fig. 2-4. The total volume bounded by this surface and the xy plane is equal to 1 in accordance with Property 2 above. The probability that X lies between a and b while Y lies between c and d is given graphically by the shaded volume of Fig. 2-4 and mathematically by

$$P(a < X < b, c < Y < d) = \int_{x=a}^b \int_{y=c}^d f(x, y) dx dy$$

The joint distribution function of X and Y in this case is defined by

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{u=-\infty}^x \int_{v=-\infty}^y f(u, v) du dv$$

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y)$$

i.e., the density function is obtained by differentiating the distribution function with respect to x and y .

$$P(X \leq x) = F_1(x) = \int_{u=-\infty}^x \int_{v=-\infty}^y f(u, v) du dv$$

We call (24) and (25) the marginal distribution functions, or simply the distribution functions, of X and Y , respectively. The derivatives of (24) and (25) with respect to x and y are then called the marginal density functions, or simply the density functions, of X and Y and are given by

$$f_1(x) = \int_{v=-\infty}^{\infty} f(x, v) dv$$

$$f_2(y) = \int_{u=-\infty}^{\infty} f(u, y) du$$

Independent Random Variables

Suppose that X and Y are discrete random variables. If the events $X = x$ and $Y = y$ are independent events for all x and y , then we say that X and Y are independent random variables. In such case,

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

$$\text{or equivalently } F(x, y) = F_1(x)F_2(y)$$

where $F_1(z)$ and $F_2(y)$ are the (marginal) distribution functions of X and Y , respectively. Conversely, X and Y are independent random variables if for all x and y , their joint distribution function $F(x, y)$ can be expressed as a product of a function of x alone and a function of y alone (which are the marginal distributions of X and Y , respectively). If, however, $F(x, y)$ cannot be so expressed, then X and Y are dependent. For continuous independent random variables, it is also true that the joint density function $f(x, y)$ is the product of a function of x alone, $f_1(x)$, and a function of y alone, $f_2(y)$, and these are the (marginal) density functions of X and Y , respectively

References:

Engineering Mathematics, T Veerarajan, TMH

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

EXPECTATION

LECTURE NO: -3

Expectation:

The expectation of a random variable X is defined as

$$E(X) = \bar{X} = \begin{cases} \sum_i x_i p_i & \text{if } X \text{ is discrete random variable with pmf } p_i \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } x \text{ is continuous random variables} \end{cases}$$

Q.1 Find the expectation of the number on a dice when thrown.

Sol: Let X be the random variable which represents the number on dice when thrown, then its probability distribution is

X	1	2	3	4	5	6
P(X)	1/6	1/6	1/6	1/6	1/6	1/6

$$E(X) = 1/6(1+2+3+4+5+6) = 7/2 = 3.5$$

Q.2 Two unbiased dice are thrown. Find the expected value of the sum of number of points on them.

Sol: Let X denotes the sum of number of points obtained on throwing two dice. Its probability distribution is

X	2	3	4	5	6	7	8	9	10	11	12
P(X)	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

$$\text{Hence } E(X) = \sum_i p_i x_i$$

$$E(X) = 2(1/36) + 3(2/36) + 4(3/36) + 5(4/36) + 6(5/36) + 7(6/36) + 8(5/36) + 9(4/36) + 10(3/36) + 11(2/36) + 12(1/36) = (1/36)252 = 7$$

Addition Theorem of Expectation:

The mathematics expectation of the sum of n random variables is equal to the sum of their expectations.

Symbolically if $x_1, x_2, \dots, x_n$ are random variables then

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$\text{Or } E\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n E(x_i)$$

Provided all expectations exist.

In particular if $n=2$, we have

$$E(x_1+x_2)=E(x_1)+E(x_2)$$

Multiplication Theorem of Expectation:

The mathematical expectation of the product of a number of independent random variables is equal to the product of their expectations. Symbolically if $x_1, x_2, \dots, x_n$ are n independent random variables then

$$E\left(\prod_{i=1}^n (x_i)\right) = \prod_{i=1}^n E(x_i)$$

Provided all expectations are exists.

In particular, if $n=2$, then

$$E(x_1 x_2) = E(x_1) E(x_2)$$

If x is random variable and a and b are constants then

$$E(ax+b) = aE(x)+b$$

Provided all the expectations exist.

1. What is expectation of the (i) sum of the points (ii) product of the points obtained in a random throw of n dice?

Sol: Let x_i denote the number obtained on throw of the i th dice, then

- (i) Sum of the points

$$E\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n E(x_i) = \sum_{i=1}^n \left(\frac{7}{2}\right) = \left(\frac{7}{2}\right) \sum_{i=1}^n 1 = 7n/2$$

- (ii) Product of Points

$$E(x_1 x_2 \dots x_n) = E(x_1) E(x_2) \dots E(x_n)$$

$$\left(\frac{7}{2}\right) \left(\frac{7}{2}\right) \dots n \text{ times} = \left(\frac{7}{2}\right)^n$$

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

MOMENTS

LECTURE NO: -4

Moments:

Moments in statistics is analogous to the same term in statics according to which moment is the tendency of a force to rotate a body about a point. The concept of moment for a random variable X is also extended to the higher powers of the variable X of given frequency distribution.

Moment about origin:

Let X be a discrete random variable with probability mass function $P(X=x_i)=p_i$, then r^{th} order moment about origin is defined as

$$\mu'_r = E(X^r) = \sum_i x_i^r p_i$$

If instead of pmf we are given the corresponding frequency distribution x_i/f_i , $i=1,2,\dots,n$, then r^{th} order moment is defined as

$$\mu'_r = \frac{\sum_i x_i^r f_i}{\sum_i f_i}$$

If X is a continuous random variable with pmf $f(x)$ then

$$\mu'_r = \int_{-\infty}^{\infty} x^r f(x) dx$$

In particular for $r=1$,

$$\mu'_1 = \sum_i x_i p_i = \int_{-\infty}^{\infty} x f(x) dx, \text{ which is mean.}$$

Hence mean is first order moment about origin.

Also $r=0$ $\mu'_0 = 1$ and $\mu'_1 = \bar{X} = \mu$

Moment about Mean or Central Moments:

Let X be a discrete random variable with probability mass function $P(X=x_i)=p_i$, then r^{th} order central moment is defined as

$$\mu_r = E[(X - \bar{X})^r] = \sum_i (x_i - \bar{X})^r p_i$$

If instead of probability mass function we are given the corresponding frequency distribution $\frac{x_i}{f_i}$, $i = 1,2,3,\dots,n$. then r^{th} order central moment is defined as

$$\mu_r = \frac{\sum_i (x_i - \bar{X})^r f_i}{\sum_i f_i}$$

If X is a continuous random variable with pmf $f(x)$ then

$$\mu_r = \int_{-\infty}^{\infty} (x_i - \bar{X})^r f(x) dx$$

In particular, $r=0$ then $\mu_0 = 1$

$r=1$ then $\mu_1 = 0$

$r=2$ then $\mu_2 = \sum_i (x_i - \bar{X})^2 p_i = \sigma^2 = \text{var}(X)$

we now find the intersection between moment about origin and moment about mean, considering X as discrete random variable, which would be valid in all cases

now $\mu_2 = \mu_2' - \mu_1'^2$

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2\mu_1'^3$$

Similarly we get

$$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6\mu_1'^2\mu_2' - 3\mu_1'^4 \text{ and so on.}$$

Moment about any arbitrary point:

Let X be a discrete random variable with pmf $P(X=x_i)=p_i$ then r^{th} order moment about the point a is defined as

$$\mu_r'' = E[(X - a)^r] = \sum_i (x_i - a)^r p_i$$

If instead of probability distribution we are given the corresponding frequency distribution $\frac{x_i}{f_i}, i=1,2,\dots,n$, the r^{th} order moment about the point $x=a$ is defined as

$$\mu_r'' = \frac{\sum_i (x_i - a)^r f_i}{\sum_i f_i}$$

If X is a continuous random variable with pdf $f(x)$ then

$$\mu_r'' = \int_{-\infty}^{\infty} (x_i - a)^r f(x) dx$$

In particular, $r=0$ then $\mu_0'' = 1$

$r=1$ then $\mu_1'' = \bar{X} - a$

now we find the interrelation between moment about the $x=a(\mu_r'')$ and the moment about mean (μ_r)

$$\mu_2 = \mu_2'' - \mu_1''^2$$

$$\mu_3 = \mu_3'' - 3\mu_2''\mu_1'' + 2(\mu_1'')^3$$

$$\mu_4 = \mu_4'' - 4\mu_3''\mu_1'' + 6\mu_2''\mu_1''^2 - 3\mu_1''^4 \text{ and so on.}$$

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

MOMENTS GENERATING FUNCTION

LECTURE NO: -5

MOMENTS GENERATING FUNCTION:

The moments of most distribution can be determined directly by evaluating the necessary integrals or sums. The moment generating function, though has some limitations, helps us to find moment without evaluating integral or sum.

The moment generating function (mgf) of a random variable X having the probability function $f(x)$, is given by :

$$M_X(t) = E(e^{tx}) = \sum_x e^{tx} f(x) \text{ for discrete RV}$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx \text{ for continuous RV}$$

Provided the right hand side is absolutely convergent for some positive number h such that $-h < x < h$ where t is any real parameter.

$$\begin{aligned} \text{Now } M_X(t) &= E(e^{tx}) = E\left[1 + tX + \frac{t^2 X^2}{2!} + \dots + \frac{t^r X^r}{r!} + \dots\right] \\ &= 1 + tE(X) + \frac{t^2}{2!} E(X^2) + \dots + \frac{t^r}{r!} E(X^r) + \dots \\ &= 1 + \mu'_1 t + \mu'_2 \frac{t^2}{2!} + \dots + \mu'_r \frac{t^r}{r!} + \dots \end{aligned}$$

Where μ'_r is the r th order moment about origin. Since $M_X(t)$ generates moments, hence it is known as moment generating function.

$$\text{Also } \mu'_r = \left[\frac{d^r}{dt^r} M_X(t) \right]_{t=0}$$

Similarly we can define Moment generating function about mean ($\bar{X}$) which generates central moments.

$$\begin{aligned} M_X(t) (\text{about } \bar{X}) &= E(e^{t(X-\bar{X})}) \\ &= E\left[1 + t(X-\bar{X}) + \frac{t^2}{2!} (X-\bar{X})^2 + \dots + \frac{t^r}{r!} (X-\bar{X})^r + \dots\right] \\ &= 1 + tE(X-\bar{X}) + \frac{t^2}{2!} E(X-\bar{X})^2 + \dots + \frac{t^r}{r!} E(X-\bar{X})^r + \dots \\ &= 1 + \mu_1 t + \mu_2 \frac{t^2}{2!} + \dots + \mu_r \frac{t^r}{r!} + \dots \end{aligned}$$

$$\text{Hence } \mu_r = \left[\frac{d^r}{dt^r} M_X(t) (\text{about } \bar{X}) \right]$$

In general, Moment generating function about the point $X=a$ is

$$M_X(\text{about } X=a) = E(e^{t(X-a)}) = 1 + \mu_1'' t + \mu_2'' \frac{t^2}{2!} + \dots + \mu_r'' \frac{t^r}{r!} + \dots$$

Example 60. Let the random variable X assume the value ' r ' with the probability law $P(X = r) = q^{r-1} p$, $r = 1, 2, 3, \dots$. Find the mgf of X and hence its mean and variance. Verify it by finding the mean from usual definition.

Solution : Mgf of $X = M_X(t) = E(e^{tX}) = \sum_{r=1}^{\infty} e^{tr} P_r$

$$= \sum_{r=1}^{\infty} e^{tr} q^{r-1} p = \frac{p}{q} \sum_{r=1}^{\infty} (qe^t)^r$$

$$\begin{aligned}
&= \frac{p}{q} [qe^t + (qe^t)^2 + (qe^t)^3 + \dots] \\
&= \frac{p}{q} \cdot qe^t [1 + qe^t + (qe^t)^2 + \dots] \\
&= pe^t \cdot \frac{1}{1 - qe^t} = \frac{pe^t}{1 - qe^t}
\end{aligned}$$

Now $\mu'_1 = \text{mean} = \left[\frac{dM_x(t)}{dt} \right]_{t=0} = p \left[\frac{(1 - qe^t)e^t - e^t(-qe^t)}{(1 - qe^t)^2} \right]_{t=0}$

$$= \left[\frac{pe^t}{(1 - qe^t)^2} \right]_{t=0} = \frac{p}{(1 - q)^2} = \frac{p}{p^2} = p$$

and $\mu'_2 = \left[\frac{d^2 M_x(t)}{dt^2} \right]_{t=0} = \left(p \frac{[(1 - qe^t)^2]e^t - 2e^t(1 - qe^t)(-qe^t)}{(1 - qe^t)^4} \right)_{t=0}$

$$= \left(p(1 - qe^t) \frac{e^t [(1 - qe^t) + 2e^t q]}{(1 - qe^t)^4} \right)_{t=0}$$

$$= \left[\frac{pe^t(1 + qe^t)}{(1 - qe^t)^3} \right]_{t=0} = p \frac{(1 + q)}{(1 - q)^3} = \frac{p(1 + q)}{p^3} = \frac{1 + q}{p^2}$$

Also $\text{Var}(X) = \mu'_2 - \mu_1^2 = \frac{1 + q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$.

By usual definition :-

$$E(X) = \sum_{i=1}^{\infty} i P(X=i) = \sum_{i=1}^{\infty} i (q^{i-1} p) = p [1 + 2q + 3q^2 + \dots]$$

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

MEAN AND VARIANCE

LECTURE NO: -6

Mean and Variance:

Let X be a discrete random variable with pmf $P(X=x_i)=p_i$ then variance of X is defined as

$$\text{Var}(X)=\sigma^2=E(|X_i-\bar{X}|^2)$$

Where $\bar{X}$ is the mean of random variable X .

$$\text{Now } \text{Var}(X)=\sigma^2=E(|X_i-\bar{X}|^2)=\sum_i (x_i - \bar{X})^2 p_i$$

$$=\sum_i x_i^2 p_i + 2\bar{X} \sum_i x_i p_i + \bar{X}^2 \quad (\text{as } \sum p_i = 1)$$

$$=\sum_i x_i^2 p_i - 2\bar{X}^2 + \bar{X}^2$$

$$=\sum_i x_i^2 p_i - \bar{X}^2$$

$$\sigma^2=E(X^2)-[E(X)]^2$$

This simplified formula holds equally good for continuous random variables. Hence if X is a continuous random variable

$$\sigma^2 = \int_{-\infty}^{\infty} (x_i - \bar{X})^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - [\int_{-\infty}^{\infty} x f(x) dx]^2$$

$$\text{Hence } \text{Var}(aX)=a^2\text{Var}(X)$$

Example 43. Two cards are drawn successively with replacement from a well shuffled pack of 52 cards, find the mean and variance of number of aces.

Solution : Let random variable X denote the number of aces. Hence $X = 0, 1, 2$.

$$P(\text{drawing an ace}) = \frac{4}{52} = \frac{1}{13}; \quad P(\text{not drawing an ace}) = \frac{12}{13}$$

$$\therefore P(X=0) = \frac{12}{13} \times \frac{12}{13} = \frac{144}{169}$$

$$P(X=1) = \frac{1}{13} \times \frac{12}{13} + \frac{12}{13} \times \frac{1}{13} = \frac{24}{169}$$

$$P(X=2) = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$$

Hence the required distribution is :-

X	:	0	1	2
P(X = x)	:	$\frac{144}{169}$	$\frac{24}{169}$	$\frac{1}{169}$

$$\text{Now } E(X) = \frac{24}{169} + \frac{2}{169} = \frac{26}{169}$$

$$E(X^2) = \frac{24}{169} + \frac{4}{169} = \frac{28}{169}$$

$$\begin{aligned} \therefore \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{28}{169} - \left(\frac{26}{169}\right)^2 = \frac{4732 - 676}{169 \times 169} = \frac{4056}{169 \times 169} \\ &= \frac{24}{169} \end{aligned}$$

$$\text{Hence } E(X) = \frac{26}{169} \text{ and } \text{Var}X = \frac{24}{169}$$

$$\Rightarrow \text{Standard deviation} = \sqrt{\text{Var}X} = \frac{2\sqrt{6}}{13}$$

Example 44. A random variable X has the following probability distribution :-

x_i	-2	-1	0	1	2	3
p_i	0.1	k	0.2	2k	0.3	k

- (i) Calculate the mean of X.
- (ii) Variance of X.

Solution : As above given is probability distribution hence $\sum_i p_i = 1 \Rightarrow 0.6 + 4k = 1 \Rightarrow k = 0.1$

Hence we construct the following :

x_i	p_i	$p_i x_i$	$p_i x_i^2$
-2	0.1	-0.2	0.4
-1	0.1	-0.1	0.1
0	0.2	0	0
1	0.2	0.2	0.2
2	0.3	0.6	1.2
3	0.1	0.3	0.9
Total		$\sum p_i x_i = 0.8$	$\sum p_i x_i^2 = 2.8$

$$\therefore E(X) = \sum p_i x_i = 0.8$$

$$E(X^2) = \sum p_i x_i^2 = 2.8$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= (2.8) - (0.8)^2 = 2.16 \end{aligned}$$